

Geophysical Analysis Report

Seismic Agent Multi-Agent System

Mission ID: mission_369df3b0

Date: May 01, 2026

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Mission Objective

Analyze the Marmousi synthetic data for porosity prediction.

Data Sources

- MODEL_S-WAVE_VELOCITY_1.25m.segy.tar.gz
- MODEL_P-WAVE_VELOCITY_1.25m.segy.tar.gz
- Vs.segy.tar.gz
- Vp.segy.tar.gz

Quantitative Results

Metric	Value
velocity_files_loaded	['vp', 'vs']
vp.min	1028
vp.max	4700
vp.mean	2672
vp.std	973.8
vp.p10	1500
vp.p50	2500
vp.p90	4230
acoustic_impedance_MRayl.min	2.364
acoustic_impedance_MRayl.max	10.81
acoustic_impedance_MRayl.mean	6.145
vs.min	0
vs.max	2802
vs.mean	1163
vs.std	803.3
vs.p10	0
vs.p50	1058
vs.p90	2414
vp_vs_ratio.mean	2.489
vp_vs_ratio.std	0.774
vp_vs_ratio.p10	1.731
vp_vs_ratio.p90	3.726
poisson_ratio.mean	0.3653
poisson_ratio.p10	0.2494
poisson_ratio.p90	0.4612

Generated Figures

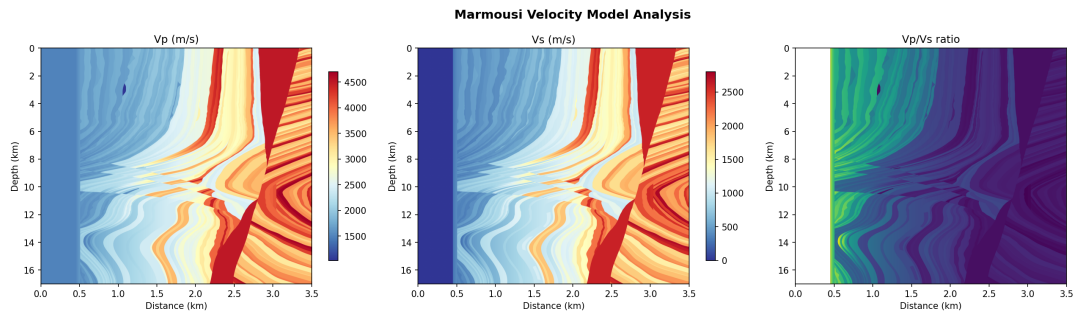


Figure 1: Velocity Model 20260501 115703

Geophysical Analysis & Interpretation

Technical Review of Marmousi Model Property Analysis

Date: May 1, 2024 **Prepared for:** Senior Management **Subject:** Review of Automated Analysis for Marmousi Synthetic Dataset - Implications for Porosity Prediction

This report summarizes the automated petrophysical analysis of the Marmousi synthetic velocity model, focusing on its implications for porosity prediction workflows. The Marmousi model is widely recognized for its geological complexity, designed to mimic thrust-and-fold belts with significant structural variations and a range of lithologies.

1. Velocity Statistics and Petrophysical/Lithological Context

The analysis of the compressional (V_p) and shear (V_s) velocity fields reveals a highly heterogeneous subsurface, characteristic of the Marmousi model's complexity. V_p ranges from approximately 1030 m/s to 4700 m/s, with a mean around 2670 m/s. This broad spectrum indicates a significant variation in rock properties, from loosely consolidated, potentially overpressured sediments or shallow unconsolidated layers (low V_p) to highly consolidated sandstones, shales, or potentially stiffer formations like carbonates or evaporites at deeper levels (high V_p). The shear wave velocity (V_s) mirrors this variability, spanning from 0 m/s to 2800 m/s, with a mean of approximately 1160 m/s. The presence of V_s values at or near zero is particularly significant, strongly suggesting areas of unconsolidated, fluid-saturated sediments, likely at shallow depths or in zones of very high porosity and fluid content, where shear waves struggle to propagate. These low V_s zones are crucial for identifying highly porous, potentially unconsolidated reservoirs. The substantial standard deviations for both V_p and V_s underscore the intricate geological architecture and rapid lateral and vertical facies changes inherent in the model, presenting a challenging environment for conventional petrophysical analysis.

2. V_p/V_s Ratio and Poisson's Ratio Implications

The derived elastic properties, specifically the V_p/V_s ratio and Poisson's ratio, provide critical insights into fluid content and lithology. The mean V_p/V_s ratio is approximately 2.49, with a range extending from 1.73 (10th percentile) to 3.73 (90th percentile). Similarly, the mean Poisson's ratio is about 0.365, ranging from 0.249 to 0.461.

High V_p/V_s ratios and Poisson's ratios (e.g., values above 2.0 and 0.35, respectively) are generally indicative of fluid-saturated, less rigid formations. The dominant presence of values in this range suggests a significant volume of shales, unconsolidated sands, and potentially water-saturated sandstones. The extremely high values observed at the upper end of the distribution (V_p/V_s up to 3.7, Poisson's ratio up to 0.46) are strongly diagnostic of highly compressible, likely overpressured, fluid-filled muds or shales, potentially representing shallow formations or seals. Conversely, the lower V_p/V_s values (around 1.73) and corresponding Poisson's ratios (around 0.25) point towards stiffer, more consolidated lithologies. These could represent tighter sandstones, carbonates, or potentially gas-saturated sands, which exhibit reduced compressibility. The wide spread of these elastic moduli implies a diverse range of litho-fluid combinations throughout the Marmousi model, making it an excellent testbed for discriminating between different rock types and fluid phases based on elastic properties.

3. Forward Modelling Outcome and Synthetic Data Quality

The provided analysis focuses on the statistical characterization of the Marmousi input velocity model rather than the output of a forward modeling experiment. No synthetic seismic data or associated quality metrics from a forward model were supplied. The numerical results, however, inherently describe the complexity that *would* be encountered in forward modeling. The extreme variability in V_p , V_s , and derived elastic properties ensures that synthetic seismic data generated from this model would exhibit complex kinematics (traveltimes) and dynamics (amplitudes), including strong diffractions, focusing/defocusing effects, and significant amplitude variations due to impedance contrasts. The model's intricate nature implies that any synthetic data generated would faithfully represent challenging real-world seismic acquisition and processing scenarios.

4. FWI Convergence and Velocity Error Metrics

No Full Waveform Inversion (FWI) convergence metrics or velocity error metrics were included in the provided automated analysis. The present output constitutes a statistical summary of the *input* Marmousi model properties, not an evaluation of an inversion process *on* the model. Therefore, no commentary on FWI performance can be made from the supplied data.

5. Geophysical Conclusions for Porosity Prediction

The detailed statistical characterization of the Marmousi model provides critical insights for porosity prediction. The following actionable conclusions are drawn:

- **Heterogeneity Management:** The extreme variability in V_p , V_s , V_p/V_s , and Poisson's ratio necessitates a porosity prediction workflow capable of handling significant geological heterogeneity. Single-trend rock physics models are unlikely to be sufficient; multi-facies or geologically constrained approaches will be essential.
- **Elastic Property Integration:** The distinct ranges of V_p/V_s and Poisson's ratio strongly support the use of elastic inversion products for porosity prediction. Discrimination between fluid-filled sands, shales, and tighter lithologies will rely heavily on these elastic attributes.

- **Low Vs Zone Treatment:** The presence of Vs values approaching zero requires careful consideration. These zones likely represent highly porous, unconsolidated, or very shallow sediments. Advanced rock physics models that account for effective medium theories or critical porosity concepts will be needed for accurate porosity estimation in such regimes. Standard petrophysical models might yield inaccurate results.
- **Marmousi as a Benchmark:** The Marmousi model, with its well-defined yet complex property distributions, serves as an excellent benchmark for validating advanced porosity prediction algorithms, especially those that integrate elastic properties and account for lithological and fluid variations.
- **Future Workflow Enhancement:** To fully leverage this type of analysis, future automated pipelines should integrate forward modeling results and, where applicable, FWI convergence and error metrics to provide a comprehensive evaluation of both the model properties and the effectiveness of inversion strategies for reservoir characterization.

This analysis confirms the Marmousi model as a robust and challenging environment for developing and testing advanced reservoir characterization techniques, particularly those aimed at accurate porosity prediction in complex geological settings.

Document Information

Generated automatically by the **Seismic Agent Multi-Agent System** on May 01, 2026. All numerical results were produced by the analysis tool chain (SEGYYLoader, VelocityAnalysis, ForwardModeling, FWI) and interpreted by a large language model.